AnchorRAG: Trust-Aware Evidence Retrieval and Anchor-Guided Synthesis for Controversial Question Answering

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Abstract

Retrieval-augmented generation (RAG) systems are helpful for controversial question answering (QA) by going beyond topical relevance and supporting answers with trustworthy, diverse, and well-grounded evidence. However, existing retrieval pipelines still prioritize semantic similarity, which can surface redundant, low-trust, or weakly supporting evidence while leaving the generator with an unstable basis for multi-perspective synthesis. In this work, we propose AnchorRAG, a framework that combines trust-aware evidence retrieval with anchor-guided answer synthesis for controversial QA. At the retrieval stage, AnchorRAG reranks candidate evidence with a trust-sensitive objective while preserving coverage across *discourse communities*, clusters of articles that share a sub-issue or argumentative stance recovered from a topic-specific evidence graph, so that the retrieved set spans the major argumentative neighborhoods rather than collapsing onto a single sub-issue. At the generation stage, it exposes retrieved evidence through structured evidence cards and selects one *community anchor* per cluster as the citeable structural backbone of the answer. We evaluate the method on seven controversial AllSides/Qbias topics with 210 open-ended queries and externally validate the source-level trust prior against Media Bias/Fact Check (MBFC) factuality labels (weighted Spearman = 0.728). Across this setting, AnchorRAG consistently improves retrieval-side trust metrics under fixed retrieval budgets and increases citation coverage, high-trust citation behavior, and backbone-trust usage in final answers. Although general answer-quality metrics that reward completeness and decisiveness show the expected trade-off, these results support our central claim: controversial QA benefits from responsibly retrieving and using stronger evidence, not from maximizing generic fluency.

CCS Concepts

• **Information systems** → **Question answering**; *Retrieval models and ranking*; • **Computing methodologies** → *Natural language generation*; • **Applied computing** → Document analysis.

Keywords

retrieval-augmented generation, controversial question answering, trustworthy retrieval, evidence synthesis, multi-perspective question answering

1 Introduction

Retrieval-augmented generation (RAG) has become a standard strategy for grounding large language models in external knowledge [16, 19]. By retrieving supporting documents and conditioning generation on them, RAG systems can produce answers that are more informative, more current, and more interpretable than purely parametric models. This retrieve-then-generate pattern has been especially effective for open-domain and knowledge-intensive question answering (QA) [7, 12, 18, 29] as well as long-form response generation [5].

In controversial QA, however, topical relevance alone is not enough. It is required to retrieve evidence that is not only relevant to the query but also trustworthy, diverse, and suitable for constructing a grounded answer that accounts for competing perspectives. This requirement is especially important for politically or socially contentious questions, where evidence differs not only in content, but also in source reliability, framing, and argumentative role [1, 3, 26, 30, 33]. In such settings, a retrieval strategy based only on semantic similarity can return redundant passages, low-trust sources, or evidence that is poorly positioned to support a coherent multi-perspective response.

Prior work has started to address the weaknesses of standard RAG from several directions. Retrieval-aware generation methods expose retriever-side signals to the generator [35], while self-reflective RAG methods attempt to reduce hallucination by improving when and how retrieved evidence is used [2, 32]. Citation-aware methods improve grounding and attribution at the sentence level [34]. Structure-aware approaches incorporate graph-level context or neighborhood information rather than treating passages as isolated chunks [15, 37]. These advances are meaningful, but they do not directly solve the problem we focus on here: *controversial QA*, where the system should prioritize more trustworthy evidence while also organizing evidence across multiple discourse communities.

This paper reformulates controversial QA as a *trust-sensitive evidence organization* problem: instead of optimizing only the relevance of retrieved documents, the system also optimizes the source trust of each retrieved item and the structural distribution of the retrieved set across argumentatively distinct sub-issues. Our goal is not to maximize the utility of generic answers in isolation, but to improve the selection, organization, and use of trustworthy evidence when answers synthesize competing perspectives. Our core hypothesis is that answer quality in this setting depends not only on whether the retriever finds relevant documents, but also on whether it surfaces trustworthy backbone evidence that the generator can use as the structural core of the response. To test this hypothesis, we propose AnchorRAG, a framework that combines trust-aware evidence retrieval with anchor-guided synthesis. The retriever ranks candidate documents using a trust-sensitive objective that jointly weighs relevance, source trust, and coverage across *discourse communities*, clusters of articles that share a sub-issue, recurring argument, or local pattern of agreement and disagreement (recovered from a topic-specific evidence graph by Louvain community detection), so that the retrieved set spans the major argumentative neighborhoods rather than collapsing onto a single sub-issue. The generator then receives structured evidence cards and one *community anchor* per cluster (the highest trust-weighted relevance representative), allowing it to separate central support from secondary or conflicting claims.

Our design is motivated by a simple principle: controversial QA does not require mechanically equal treatment of all viewpoints; instead, it requires grounded multi-perspective synthesis in which disagreement is preserved when supported, but stronger and more trustworthy evidence is given greater weight. AnchorRAG therefore strengthens the connection between retrieval quality and answer construction by explicitly modeling both source trust and discourse structure.

We evaluate AnchorRAG in a reference-free setting that separates retrieval quality, evidence utilization, and answer quality, following recent efforts to evaluate RAG systems beyond final-answer utility alone [9, 31, 36]. Accordingly, AnchorRAG does not aim to universally maximize generic answer quality; rather, it tests whether controversial QA benefits from trust-sensitive evidence organization by improving retrieval trust and high-trust citation behavior, while making the trade-off with sentence-level support metrics explicit. Across seven controversial topics and 210 evaluation queries, AnchorRAG consistently improves retrieval-side trust metrics and increases the use of high-trust evidence in the final answer. Although general answer-quality metrics that reward completeness and decisiveness show the expected trade-off, this pattern is consistent with the paper's central claim: controversial QA benefits from responsibly retrieving and using stronger evidence, not from maximizing generic fluency.

The main contributions of this paper are as follows:

- **Trust-sensitive reformulation.** We reformulate controversial QA as a trust-sensitive evidence organization problem in which source trust is a first-class component of the retrieval objective (Eq. (1)).
- **AnchorRAG framework.** We propose AnchorRAG, a RAG framework that integrates trust-aware retrieval, structured evidence cards (Eq. (2)), and a single per-community anchor (Eq. (3)) to organize multi-perspective evidence around high-trust backbone support.
- **AllSides/Qbias evaluation with MBFC external validation.** We present an AllSides/Qbias evaluation over seven topics and 210 controversial queries with external validation of the source-level trust prior against Media Bias/Fact Check (MBFC) factual-reporting labels (Spearman = 0.662, weighted Spearman = 0.728 on the top-50 sources), and report 7/7 per-topic dominance on three trust-sensitive metrics (paired Wilcoxon $p < 0.01$ against all six comparison methods).

Paper organization. Section 2 surveys related work; Section 3 positions AnchorRAG against prior and controlled comparison methods; Section 4 describes the AnchorRAG framework, including the three-axis retrieval objective (Eq. (1)), the evidence-card weight (Eq. (2)), and the community-anchor score (Eq. (3)); Section 5 reports the experimental setup, main results, and ablation; Sections 6 and 7 discuss findings and conclude.

2 Related Work

2.1 Retrieval-Augmented Generation

Retrieval-augmented generation (RAG) has become a core framework for grounding large language models in external evidence. Early RAG approaches combine parametric generation with non-parametric memory, allowing models to retrieve supporting passages at inference time and generate answers conditioned on them [19]. Fusion-in-Decoder (FiD) further improved multi-passage reasoning by jointly aggregating information from multiple retrieved passages during decoding [16]. Maximal Marginal Relevance (MMR) [6] is the classical relevance–redundancy reranker for retrieval and remains a standard diversity baseline; it reduces redundancy in the retrieved set but does not model source-level reliability or community structure. More recent self-reflective and adaptive RAG methods improve when and how retrieval is used during generation [2, 17], but they decide *when* or *how* to retrieve rather than *which* sources to trust, treating all retrieved passages as equally credible. These methods established the general retrieve-then-generate paradigm, but they primarily optimize for relevance or factuality rather than explicitly modeling source-level evidence trustworthiness in controversial QA.

2.2 Retrieval- and Citation-Aware Generation

Recent work has attempted to reduce the disconnect between retrieved evidence and generated responses. ALCE frames citation generation as an end-to-end retrieval-and-generation problem and evaluates fluency, correctness, and citation quality [11]. R2AG incorporates retrieval-side information into generation in order to narrow the gap between retrievers and generators [35]. ReClaim improves attribution by interleaving reference generation and claim generation, enabling more fine-grained citation behavior in RAG outputs [34]. These methods are closely related to our goal of improving evidence utilization, but they do not directly model trust-aware evidence prioritization or the organization of evidence around discourse communities and anchors.

2.3 Graph- and Structure-Aware Retrieval

Another line of work augments RAG with structure beyond isolated passages. GRAG retrieves textual subgraphs and incorporates graph context into generation, demonstrating the value of graph-aware retrieval and generation for networked documents [15]. KG^2RAG uses knowledge graphs to guide chunk expansion and organization, improving diversity and coherence in the retrieved context [37]. Most directly related, GraphRAG [8] applies hierarchical community detection over an entity-relation graph and uses each community's *summary* as a retrieval unit for query-focused summarization; we contrast AnchorRAG's single-article-anchor design with GraphRAG's summary-unit design in detail in Section 3. Together, these methods show that structural relations among retrieved units can improve generation. Our work focuses on controversial QA over source-perspective-labeled news evidence and combines structure-aware retrieval with explicit trust-sensitive ranking and anchor-guided synthesis.

2.4 Multi-Perspective QA and Trustworthy Evidence

Our problem setting is also connected to prior work on opinion-oriented and multi-perspective question answering (QA). Earlier work on the OpQA corpus showed that opinion questions differ fundamentally from factoid QA and require synthesis across competing answers and viewpoints [33]. In news settings, this challenge is compounded by variation in media reliability and ideological bias. Prior work has examined how to predict factuality and bias of news media sources [3], and recent surveys have highlighted the importance of modeling both source reliability and political leaning in news analysis [26]. Source-trust propagation itself has a long lineage in web information retrieval; TrustRank [13] propagated a small set of human-rated seeds over the web hyperlink graph to demote spam, but these signals have rarely been carried back into the *retrieval-time scoring* of RAG systems, where source reliability is more often handled post hoc as fact-checking or factuality prediction rather than as a structural prior over retrieval candidates. In contrast to approaches that focus only on relevance, only on viewpoint balance, or only on post-hoc factuality verification, our goal is to make a source-level trust prior a first-class component of the retrieval objective and to use it together with discourse-community structure to retrieve and synthesize evidence that is both diverse and trustworthy.

2.5 Evaluation of Grounded Generation

Recent reference-free RAG evaluation frameworks (RAGAs [9], ARES [31]) and factual-consistency / attribution resources (AlignScore, FActScore, ExpertQA [23, 25, 36]) advocate decomposing answer quality into retrieval, faithfulness, and relevance components. We follow this direction but tailor the analysis to controversial QA by explicitly separating retrieval quality, trust-sensitive evidence use, and downstream answer quality (Section 5).

3 Positioning of AnchorRAG

We position AnchorRAG with a compact matrix rather than a model-by-model limitation list. Table 1 separates general RAG capabilities—diversity-aware selection, generator-side retrieval/citation cues, and explicit graph structure—from the three trust-sensitive constraints required by our controversial-QA formulation: source-trust ranking at retrieval time, source-trust exposure in the generator's organizing structure, and citeable article-level provenance through final citations.

On the retrieval side, standard semantic RAG and MMR optimize relevance or redundancy, while graph-aware methods and our controlled graph-retrieval comparison add structural coverage. These mechanisms are useful but are orthogonal to source reliability unless the ranking objective explicitly includes a source-trust prior. AnchorRAG makes this prior a first-class term in Eq. (1), so trust affects which evidence enters the retrieved set rather than only how retrieved text is later interpreted.

On the generation side, R2AG and ReClaim are closest in spirit because they reduce the disconnect between retrieval and generation: R2AG exposes retrieval information to the prompt, and ReClaim improves claim-level attribution through interleaved reference generation. In Table 1, this is why they receive the generator-side cue mark. However, these cues are not source-trust signals and do not organize the answer around higher-trust evidence. AnchorRAG instead exposes trust scores and community anchors through structured evidence cards and answer planning, turning retrieval-side trust into a generation-side organizing cue.

Finally, graph-aware RAG variants show that structure helps organize multi-document evidence, but their structural units need not preserve article-level provenance. GRAG conditions generation on textual subgraphs, and GraphRAG retrieves community summaries; both are valuable graph-based abstractions, but the citeable unit is no longer a single source article with its original source label, perspective tag, and trust score. AnchorRAG keeps the retrieved article node as the structural and citeable unit, selecting one article-level anchor per retrieved community via Eq. (3).

Table 1: Positioning matrix for selected prior methods and controlled comparison methods, interpreted by their method-intrinsic mechanisms rather than by the shared experimental citation wrapper. The first three columns capture general RAG capabilities; the last three are the trust-sensitive constraints used by our controversial-QA formulation. Div. = diversity-aware selection; Gen. cue = generator-side retrieval/citation cue; Graph = explicit graph/community structure; Trust rank = source-trust signal in retrieval-time ranking; Gen. trust = source-trust signal exposed to the generator's organizing structure; Art. prov. = citeable article-level unit and source provenance preserved through citations.

Method	Div.	Gen. cue	Graph	Trust rank	Gen. trust	Art. prov.
vanilla RAG [19]	✗	✗	✗	✗	✗	✗
MMR RAG [6]	✓	✗	✗	✗	✗	✗
graph retrieval (no trust)	✓	✗	✓	✗	✗	✓
R2AG [35]	✗	✓	✗	✗	✗	✗
ReClaim [34]	✗	✓	✗	✗	✗	✗
GRAG [15]	✗	✓	✓	✗	✗	✗
GraphRAG [8]	✗	✗	✓	✗	✗	✗
AnchorRAG (ours)	✓	✓	✓	✓	✓	✓

Thus, Table 1 should be read as a capability map: prior methods cover important individual dimensions, whereas AnchorRAG is designed to satisfy the three trust-sensitive constraints jointly. This joint design explains why the empirical comparison in Section 5.5 focuses on trust-sensitive transfer into citations and answer backbones, not only on generic answer quality.

4 AnchorRAG

AnchorRAG operationalizes the trust-sensitive evidence organization reformulation introduced in Section 1: every retrieval-time and generation-time decision is taken jointly over (i) relevance, (ii) source trust, and (iii) discourse-community coverage, rather than over relevance alone. Figure 1 walks through the three-phase architecture. **Phase 1 (Evidence Graph Construction)** is offline: input documents are decomposed into evidence units and connected into an evidence graph by discourse edges labeled *support/refute/contextualize*; [1] Separately, each node receives these annotations, making the graph usable for trust-aware retrieval and anchor selection. The graph is then partitioned into discourse communities by Louvain community detection (sub-issue clusters that serve as the structural unit for both diversity-aware retrieval and anchor selection in subsequent phases). **Phase 2 (Trust-Aware Retrieval)** runs per query: candidate evidence is scored by a trust-based reranking objective that combines query relevance, node-level trust, and a community-diversity multiplier (the three axes of the reformulated objective), then one community anchor per

[1]The graph becomes trust-aware through node-level source-reliability and structural-importance annotations that are later used in retrieval and anchor selection.

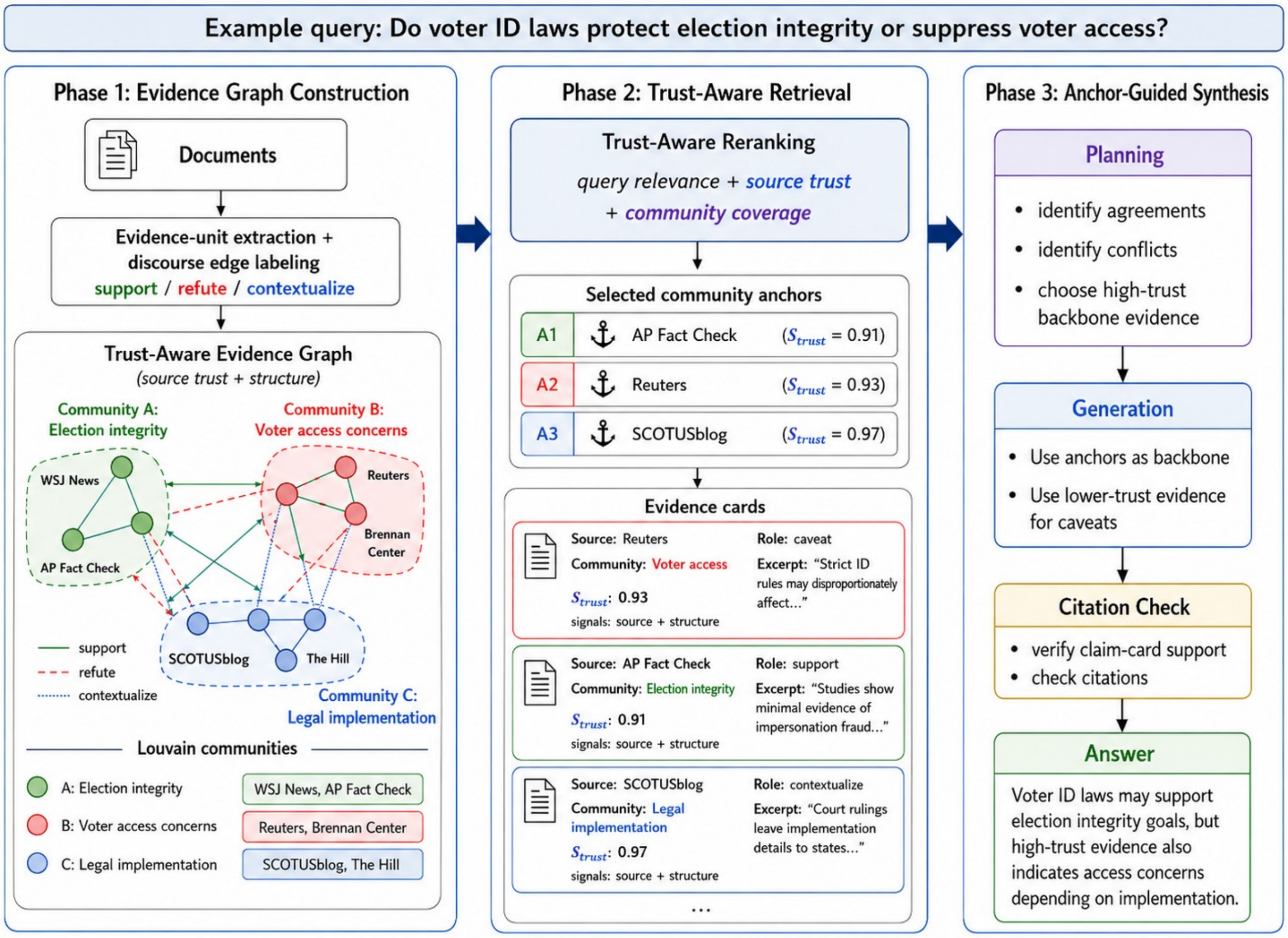

Figure 1: Overall architecture of AnchorRAG, comprising three phases: Phase 1 Evidence Graph Construction, Phase 2 Trust-Aware Retrieval, and Phase 3 Anchor-Guided Synthesis. Node colors in Phases 1 and 2 indicate community membership; anchor icons in Phase 2 mark the selected community anchor per cluster; the citation-check box denotes the evaluation-time support check applied to generated cited claims. Per-phase details are in the body of Section 4.

retrieved community is selected as the cluster's local backbone, and each retrieved item is wrapped as an evidence card exposing short evidence text, purpose, and metadata. **Phase 3 (Anchor-Guided Synthesis)** produces the answer in two generation sub-stages: a planner reads the community anchors and evidence cards to analyze agreements/conflicts and establish a synthesis plan, and a generator uses the anchors as the answer backbone, with lower-trust evidence reserved for caveats. Claim-card support is then assessed by the shared evaluation pipeline rather than by an additional generation-time verifier in the main AnchorRAG run.

4.1 Evidence Graph Construction (Phase 1)

AnchorRAG addresses controversial QA over a topic-specific evidence graph constructed from news articles with source and perspective metadata. Each graph node corresponds to one article and stores its source, title, article text, and perspective label. To connect related evidence items, we first generate article embeddings and identify high-similarity neighbors as candidate edges. We then label semantically meaningful article pairs with discourse relations such as *support* or *refute*, , or *contextualize*, yielding a graph that captures both topical relatedness and argumentative interaction. Separately, each article carries a source-reliability prior and a graph-structural score; these node-level annotations, rather than the discourse labels themselves, make the graph usable for trust-aware retrieval and high-trust local backbone selection. Finally, we apply Louvain community detection [4, 10, 27] to identify discourse clusters, which serve as the structural units used in both retrieval and downstream synthesis.

This graph is not intended as a gold discourse representation. Rather, it is an operational approximation of how evidence is organized around sub-issues, recurring arguments, and local patterns of agreement and disagreement. In controversial QA, such structure matters because the final response depends not only on which articles are retrieved, but also on how the retrieved evidence is distributed across distinct regions of the discourse.

4.2 Trust-Aware Retrieval

Given a question q, AnchorRAG first retrieves a candidate pool of semantically relevant articles and then reranks them using a trust-aware score. Throughout this section, q denotes the input question and d denotes an article/document node in the topic-specific evidence graph. This follows the common RAG practice of separating candidate retrieval from downstream evidence selection [18, 19], but changes the reranking objective to account for source reliability and graph structure. Each node is assigned a trust-sensitive prior $S_{\text{trust}}(d) \in [0, 1]$ that linearly combines two components: a source-level reliability score $r_{\text{src}}(d) \in [0, 1]$, reflecting the estimated factual-reporting quality of d's outlet (estimated by the LLM-based source auditor described in Section 5), and a structural-importance score $r_{\text{struct}}(d) \in [0, 1]$. The source-level score is obtained once per news outlet by an LLM auditor, who assigns a continuous reliability score from 0 to 1, with higher scores reserved for outlets judged to have stronger factual reporting practices; all articles from the same outlet inherit this source score. The structural component reflects d's centrality in the topic-specific evidence graph and is computed via min–max normalization of an undirected PageRank-style centrality [28]. We validate the source-level component against MBFC factual-reporting labels in Section 5.5.

During iterative top-k selection, let S_t denote the documents already selected at step t. For a candidate document d, the retrieval score is defined as

$$s_t(d, q; S_t) = \alpha \cdot \text{rel}(d, q) + \beta \cdot S_{\text{trust}}(d) \cdot m_{\text{div}}(\text{comm}(d), S_t), \quad (1)$$

where $\text{rel}(d, q)$ denotes semantic query-document relevance, $S_{\text{trust}}(d)$ denotes the node-level trust proxy, and $m_{\text{div}}(\text{comm}(d), S_t)$ is a community-level diversity multiplier applied to the trust component during selection. In our implementation, $\alpha = 0.7$, $\beta = 0.3$, and m_{div} is a single-step repeat-community penalty. Let $n_c(S_t)$ be the number of already selected documents from community c; we set $m_{\text{div}}(c, S_t) = 1$ when $n_c(S_t) = 0$ and $m_{\text{div}}(c, S_t) = \rho$ otherwise, with $\rho = 0.8$. This setting implements a relevance-gated trust objective: relevance keeps the retrieved set responsive to the question, while the trust term shifts priority among topical candidates toward more reliable evidence. The corresponding sensitivity analysis is reported in Section 5.3.

Importantly, S_{trust} is not intended as an article-level truth label. Instead, it is a semi-automated trust proxy that combines source reliability and graph structural importance. We validate the source-level component against MBFC factual-reporting labels (Figure 2), then evaluate the full node-level proxy through retrieval-side trust/coverage diagnostics and downstream citation/backbone behavior (Tables 4 and 5).

4.3 Trust-Aware Evidence Cards

Instead of passing raw retrieved documents directly to the generator, AnchorRAG converts each retrieved item into a *structured evidence card*. Each card contains the article title, source, perspective label, trust score, query relevance, community identifier, and a short evidence excerpt. This compact representation follows the motivation of retrieval-aware and citation-aware generation, where the generator benefits from explicit retrieval-side signals rather than only raw passages [34, 35]. It also reduces the burden of locating relevant evidence inside long retrieved contexts [21].

For each retrieved document d and the same input question q, we also compute a lightweight evidence weight separate from the retrieval score:

$$w(d, q) = \gamma \cdot \text{rel}(d, q) + (1 - \gamma) \cdot S_{\text{trust}}(d), \quad (2)$$

with $\gamma = 0.65$ in our implementation. The separate parameter reflects the different decision stage: Eq. (1) selects documents during retrieval, whereas Eq. (2) orders the already selected cards for planning. We keep the same relevance–trust signal family for consistency, but set γ independently because card ordering no longer controls top-k selection or community coverage. This weight serves as a heuristic cue for distinguishing backbone evidence from weaker supporting or conflicting evidence. The evidence-card design, therefore, makes the retrieval layer legible to the generator instead of forcing the model to infer evidence quality from raw text alone.

Concretely, $w(d, q)$ is exposed in each card as an evidence-weight field and is used only as a generation-side priority cue for ordering and planning around retrieved cards; it is not used to compute the retrieval ranking in Eq. (1) or any reported trust metric. Accordingly, m_{div} appears only in the retrieval-stage selection score, while Eq. (2) scores the already selected cards for generation-side ordering.

4.4 Community Anchor Selection

To summarize the retrieved evidence at the community level, AnchorRAG selects one anchor from each retrieved discourse cluster. Intuitively, a community anchor is the retrieved article that best represents the local backbone of that cluster.

For a retrieved node d, the anchor score is defined as

$$a(d) = \lambda_1 \deg^{\text{intra}}_{\text{norm}}(d) + \lambda_2 \text{rel}_{\text{norm}}(d, q) + \lambda_3 \text{trust\_support\_ratio}(d), \quad (3)$$

where the three component scores are defined as follows. Let $c = \text{comm}(d)$ denote d's community and R_c the retrieved nodes of S_t in c. The within-community degree is min–max-normalized over R_c:

$$\deg^{\text{intra}}_{\text{norm}}(d) = \frac{\deg^{\text{intra}}(d) - \min_{d' \in R_c} \deg^{\text{intra}}(d')}{\max_{d' \in R_c} \deg^{\text{intra}}(d') - \min_{d' \in R_c} \deg^{\text{intra}}(d') + \epsilon},$$

where $\deg^{\text{intra}}(d)$ counts d's edges to other nodes in c and $\epsilon = 10^{-9}$ guards against degenerate single-node communities. The relevance component $\text{rel}_{\text{norm}}(d, q)$ is the same semantic relevance score $\text{rel}(d, q)$ from Eq. (1), min–max-normalized over the retrieved nodes R_c in the same community. The trust-supported local-structure score is

$$\text{trust\_support\_ratio}(d) = \frac{|N^{\text{HT}}_{\text{supp}}(d)|}{|N_{\text{supp}}(d)| + \epsilon}, \quad (4)$$

where $N^{\text{HT}}_{\text{supp}}(d) = \{d' \in N_{\text{supp}}(d) : S_{\text{trust}}(d') \geq \tau_{\text{anchor}}\}$, $N_{\text{supp}}(d)$ is d's *support*-labeled neighborhood, $\tau_{\text{anchor}} = 0.8$, and $(\lambda_1, \lambda_2, \lambda_3) = (0.45, 0.35, 0.20)$ in our implementation. The highest-scoring node in each retrieved community is selected as that community's anchor; the single-anchor design is empirically supported by the multi-anchor ablation (Section 5.5: 0.7705 vs. 0.7927 support; 0.6139 vs. 0.6239 trust-weighted claim support).

4.5 Anchor-Guided Synthesis

AnchorRAG performs answer generation in two sub-stages. In the first sub-stage, an evidence planner reads the community anchors and trust-aware evidence cards, producing a compact synthesis plan. The plan identifies the most strongly supported points, the main disagreements, the remaining evidence gaps, and a suggested response strategy. In the second sub-stage, the generator produces the final answer conditioned on both the plan and the card set. We evaluate citation coverage and claim-card support after generation using the shared evaluation protocol described in Section 5.

The generation policy follows an evidence-weighted balance principle. The model is instructed to use community anchors as the default backbone of the answer, to prioritize evidence with higher S_{trust} and stronger query relevance, and to preserve disagreement when it is backed by non-trivial evidence. However, the model is explicitly discouraged from giving equal weight to all perspectives when the quality of evidence is clearly asymmetric. Lower-trust or conflicting evidence is therefore used primarily to explain disagreement, caveats, or uncertainty rather than to override stronger support.

This design reflects the observation that controversial QA does not require equalization of mechanical viewpoints. Instead, it requires grounded multi-perspective synthesis that remains open to disagreement while still giving greater prominence to stronger and more trustworthy evidence.

5 Experiments

5.1 Setup

Dataset and queries. We evaluate AnchorRAG on seven controversial topics from the AllSides/Qbias corpus [14]: *Immigration*, *Gun Control and Gun Rights*, *Economy and Jobs*, *Abortion*, *Free Speech*, *Voting Rights and Voter Fraud*, and *Climate Change*, with left/center/right source-perspective labels. After topic-specific preprocessing, the seven graphs contain 534, 282, 530, 279, 225, 270, and 165 articles, respectively (Table 2). For each topic, we use 30 open-ended controversial questions designed to elicit conflicting evidence, for 210 evaluation queries in total. The final 30 questions for each topic are human-curated: we used GPT-4o and Claude to propose candidate questions, then manually reviewed, edited, removed, and added questions so that the final set covers distinct controversial sub-issues, avoids near-duplicates, and elicits conflicting evidence. We do not create gold reference answers, because controversial QA often lacks a single canonical answer; instead, our evaluation is reference-free and conditions each generated answer on its retrieved evidence.

Comparison methods and ablations. For the final comparison, we evaluate AnchorRAG against two groups of comparison methods: (i) standard or prior-method baselines, including vanilla RAG [18, 19] (semantic top-k), MMR RAG [6] (diversity-aware), R2AG [35] (retrieval-feature prompting), ReClaim [34] (reference-claim generation), and GRAG [15] (textual-subgraph retrieval); and (ii) graph retrieval (no trust), a controlled diagnostic baseline that uses the same article-level evidence graph and community structure as AnchorRAG but removes the source-trust objective. For the ablation, we compare five variants: *vanilla rag*, *wo trust* ($\alpha = 1$, $\beta = 0$), *wo anchor*, *wo diversity* ($m_{\text{div}} = 1$), and the full AnchorRAG system, plus a multi-anchor variant (two anchors per community). These ablations isolate the design choices in Eq. (1) and Section 4: *wo trust* removes the trust term from retrieval while retaining diversity and anchor-guided generation; *wo anchor* keeps trust-aware retrieval and evidence cards but removes the anchor block from answer planning/generation; *wo diversity* keeps the trust term and anchor block but disables the community-repeat penalty; and the multi-anchor variant selects two anchors per retrieved community instead of one. Across all methods, we keep the corpus, query set, generator, retrieval budget, and evaluation pipeline fixed.

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Implementation. We use `text-embedding-3-small` for query and document embeddings and `gpt-4o` (temperature 0) for both source-reliability auditing and answer planning/generation; the semantic judges run with deterministic settings and fixed rubrics. Retrieval uses a candidate pool of 50 and returns top-k with $k = 10$. We set $\alpha = 0.7$ and $\beta = 0.3$ in Eq. (1); Section 5.3 summarizes parameter sensitivity for retrieval weighting, community penalty, card ordering, and anchor selection. Statistical tests on the final downstream comparison use AnchorRAG as the reference: paired bootstrap 95% CIs (10,000 resamples, fixed seed) and Wilcoxon signed-rank tests (two-sided, with tie correction); a single asterisk and a double asterisk after a AnchorRAG table value indicate $p < 0.05$ and $p < 0.01$ from pairwise tests against each of the six comparison methods, with the mark reflecting the least significant comparison.

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5.2 Retrieval-Side Validation

We first validate the retrieval-side assumptions behind AnchorRAG. Across the seven AllSides/Qbias topics, the scored evidence graphs show consistently high modularity (0.7779–0.8860), indicating well-separated local clusters; mean community purity remains low and tightly concentrated (0.3689–0.3926), reflecting the source-level perspective labels and three-way source diversity. We therefore interpret discovered communities as sub-issue-centered discourse clusters rather than stance-pure ideological blocs (see Table 2).

Table 2: Community-structure diagnostics for the seven AllSides/Qbias topic graphs. High modularity with low and tightly concentrated mean purity supports interpreting communities as sub-issue-centered discourse clusters rather than stance-pure blocs.

Topic	Nodes	Edges	Comm.	Modul.	Purity
Immigration	534	1053	56	0.8630	0.3895
Gun Control	282	550	32	0.8372	0.3901
Economy and Jobs	530	1063	51	0.8860	0.3887
Abortion	279	688	19	0.7779	0.3835
Free Speech	225	481	26	0.8177	0.3689
Voting Rights	270	658	22	0.8017	0.3926
Climate Change	165	386	13	0.8188	0.3697

Table 3: Retrieval-side validation at a fixed budget of $k = 10$, before generation. Coverage and entropy measure discourse-community spread, Avg S_{trust} measures the trust of retrieved evidence, and Avg Rel measures mean query-document relevance.

Method	Coverage	Comm. Entropy	Avg S_{trust}	Avg Rel
Semantic top-k [18, 19]	4.4000	1.2416	0.7318	**0.4230**
MMR [6]	6.3714	1.6795	0.7207	0.4025
Graph retrieval (no trust)	9.3476	2.2060	0.7293	0.3974
AnchorRAG	**9.4619**	**2.2119**	**0.7624**	0.3910

Table 3 reports the same fixed-budget comparison directly. AnchorRAG outperforms semantic top-k retrieval used in vanilla RAG [18, 19], MMR reranking [6], and controlled graph retrieval without the source-trust term over the same Louvain community graph [4]. Pure semantic retrieval remains stronger on average query-document relevance (0.4230 vs. 0.3910), and MMR directly targets redundancy reduction; the retrieval-side gain of AnchorRAG is therefore specific to the joint trust-and-community-coverage axis targeted by our objective.

5.3 Sensitivity Analysis

We summarize parameter sensitivity under the same 210-query setting, changing one parameter family at a time around the default configuration. The retrieval-weight sweep shows the expected trust–relevance frontier: the trust-heavy setting $(\alpha, \beta) = (0.5, 0.5)$ raises High-Trust Citation Rate (HT Cite) from 0.683 to 0.899 but lowers average query-document relevance (Avg Rel) from 0.409 to 0.386, whereas the relevance-heavy setting $(0.9, 0.1)$ raises Avg Rel to 0.423 but lowers HT Cite to 0.596. We therefore use $(0.7, 0.3)$ as the default balanced operating point, with $(0.5, 0.5)$ treated as a trust-heavy sensitivity setting rather than the main configuration.

The community penalty controls a different trade-off: removing the repeat-community penalty ($\rho = 1.0$) reduced community coverage from 7.54 to 4.36 and lowered HT Cite by 0.025, whereas a stronger penalty ($\rho = 0.6$) increased coverage to 9.25 but lowered Avg Rel from 0.409 to 0.395 without a commensurate HT Cite gain. We therefore use $\rho = 0.8$ as a soft repeat-community penalty rather than a hard quota; additional progressive penalties for the third and later items were not beneficial in the retrieval trust–relevance balance.

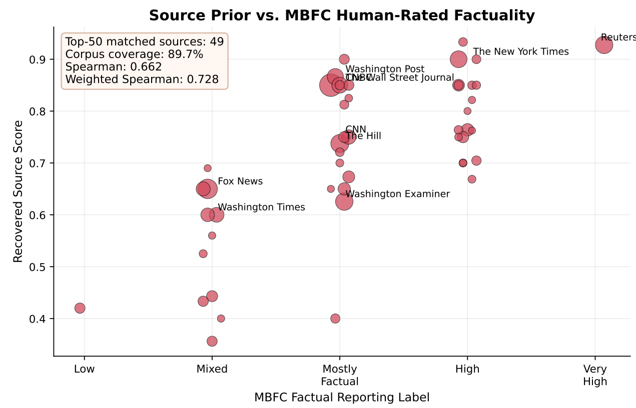

Figure 2: Validation of the source-level reliability prior against MBFC factual reporting labels on the top-50 most frequent AllSides/Qbias sources.

The generation-side parameters were more stable: varying the card-ordering weight over $\gamma \in \{0.5, 0.65, 0.8\}$ changed HT Cite by less than 0.01, changing anchor-score weights around $(\lambda_1, \lambda_2, \lambda_3) = (0.45, 0.35, 0.20)$ moved HT Cite by at most 0.02, and changing τ_{anchor} from 0.8 to 0.70 or 0.85 reduced HT Cite by about 0.02–0.03. These results suggest that the main operating frontier is governed by retrieval-time trust/relevance and community-coverage choices, while card ordering and anchor scoring mainly provide stable generation-side organization.

5.4 Validation of the Source-Level Reliability Prior

Because the trust-sensitive retrieval metrics depend on a source-level reliability prior, we compare the source-level reliability component $r_{\text{src}}(d)$ of AnchorRAG against Media Bias/Fact Check (MBFC) factual reporting labels. This score is assigned once per news outlet by the source auditor and inherited by articles from that outlet; we focus on this source-level component rather than the full node-level trust score because S_{trust} combines both source reliability and graph structural information. For the top-50 most frequent source entries in AllSides/Qbias, we successfully match 49 sources to MBFC labels, covering 1779 of 1983 documents (89.7%). The source-level reliability component r_{src} shows substantial alignment with MBFC's human-rated factual reporting labels: Spearman = 0.662, weighted Spearman = 0.728 in the top-50 setting (Figure 2). The full node-level proxy shows similar alignment, indicating that the graph-structural term does not substantially distort the source-reliability signal. Beyond this prior-level alignment, we further use the same MBFC labels as an external check on downstream citation and backbone-trust behavior in Section 5.5 (Table 5).

5.5 Main Results: Downstream Transfer Analysis

Trust transfer to citations and answer backbone. Table 4 shows that AnchorRAG improves the transfer of high-trust retrieved evidence into citations and answer backbones. Among the compared methods, AnchorRAG achieves the strongest *High-Trust Citation Rate*, *Anchor Utilization*, *Backbone Trust Rate*, and *Avg* $S_{\text{trust}}@k$. On HT Cite, Backbone Trust, and Avg $S_{\text{trust}}@k$, paired tests reject zero mean difference at $p < 0.01$ against all six comparison methods. Anchor Utilization is not significantly different because graph retrieval (no trust) has a higher per-query representative-citation rate under this post hoc definition. ReClaim obtains the highest raw *Citation Coverage* because its generation procedure explicitly interleaves references and claims, encouraging nearly every factual sentence to carry a citation. This improves citation density, but it does not change the retrieval-time source selection objective or prioritize source reliability; in our outputs, the extra citations often repeat overlapping clauses, so the higher coverage does not translate into higher HT Cite, Backbone Trust, Support, or Utility. The same transfer pattern appears on the late-term abortion query (abo-02): AnchorRAG surfaces SCOTUSblog ($S_{\text{trust}} = 0.97$) as a top-ranked card and reaches 1.00 on both *High-Trust Citation Rate* and *Backbone Trust*, whereas vanilla rag obtains 0.00 on both.

External validation under MBFC source-factuality labels. A natural concern about Table 4 is circularity: a system that retrieves and cites by its own S_{trust} may be favored on metrics derived from S_{trust}. Table 5 addresses this by re-scoring the same generated answers and inline card citations from Table 4 with MBFC factual reporting labels as an external, human-rated source-level proxy. Instead of re-evaluating every metric in Table 4, we focus on the two citation/backbone metrics that depend most directly on the internal source-trust signal. MBFC HT Cite and MBFC Backbone HT are external-label counterparts of HT Cite and Backbone Trust, computed over citations whose sources can be mapped to MBFC. The remaining MBFC columns are auxiliary diagnostics for average source factuality and source-mapping coverage, not replacements for Table 4's citation-density, anchor-utilization, or internal-S_{trust} metrics. Each inline card citation (e.g., `[Card 2]`) is mapped to its cited source's MBFC factual-numeric label (1=Low, 5=Very High) using a 72-source mapping covering 87–93% of citations across methods; citations to sources with labels ≥ 4 are counted as high-factuality. Across the five MBFC columns, AnchorRAG obtains the best value in the table: MBFC HT Cite (0.4228), MBFC Avg Factual (3.3648), MBFC Backbone HT (0.4140), MBFC Backbone Avg Factual (3.3587), and MBFC Map Cov. (0.9308). The average-factuality gains are significant at $p < 0.01$ for both all citations and backbone citations; MBFC Backbone HT and MBFC Map Cov. are significant at $p < 0.05$. The only weaker case is all-citation MBFC HT Cite, which is borderline against vanilla rag ($\Delta = +0.041$, 95% CI $[-0.001, +0.083]$, $p = 0.078$), while the corresponding backbone metric is significant against vanilla rag ($\Delta = +0.065$, $p = 0.011$).

5.6 Generated-Answer Quality and Trust–Support Trade-off

"Answer quality" refers to the final generated response, evaluated reference-free from the question, retrieved evidence, and citations. In Table 6, Groundedness measures whether the response as a whole is grounded in the retrieved cards, Evidence Support Rate checks whether cited factual sentences are supported by their cited cards, and Utility measures completeness and helpfulness for the controversial question. These diagnostics show that AnchorRAG's trust-sensitive gains come at some cost to general answer quality, but not at the expense of groundedness or support: the full model remains competitive on Evidence Support Rate while trailing stronger relevance-oriented baselines on Groundedness and Utility. The graph-retrieval control explains the main failure mode: without the source-trust objective, it can select more directly responsive evidence from the same community graph, whereas AnchorRAG prioritizes a high-trust answer backbone and can under-answer when that backbone is only indirectly responsive.

The trust-sensitive gains are not an aggregate artifact: across all seven topics, AnchorRAG obtains the highest *High-Trust Citation Rate*, and the same per-topic dominance also holds for *Backbone Trust* and *Avg* $S_{\text{trust}}@k$ (7/7 topics each; minimum per-topic gap over the best comparison method +0.13, +0.11, and +0.02 respectively). Table 7 complements these aggregate results with a diagnostic slice against graph retrieval (no trust). We use this graph control as the slice comparator because it is the closest controlled no-trust variant of AnchorRAG, sharing the same community graph while removing the source-trust objective, and it is also the strongest comparison method on Evidence Support Rate in Table 6. For Groundedness and Utility, the strongest comparison methods are vanilla/MMR semantic retrieval, so those losses are interpreted in Table 6 rather than through the graph-only slice.

We select the top 10 queries, among 210 total queries, where AnchorRAG improves most over graph retrieval (no trust) in HT Cite, subject to two constraints: citation coverage must also improve, and evidence support must not decrease. In this selected slice, average HT Cite improves by +0.8028 and Citation Coverage by +0.4272, while *Evidence Support Rate* increases from 0.7324 to 1.0000. The slice should therefore be read as query-level evidence that the aggregate trust–support trade-off is not mechanically inevitable, rather than as an unbiased estimate of overall performance. The metrics where AnchorRAG is not best follow from method objectives: ReClaim has the highest Citation Coverage because it explicitly inserts citations during generation; vanilla/MMR have the highest Groundedness and Utility because semantic retrieval favors direct and complete responses; and graph retrieval (no trust) has the highest Support because the same graph structure without the trust objective can select locally supportive evidence without shifting the backbone toward higher-trust sources. These answer-quality losses are real, but they are consistent with AnchorRAG's role as a trust-sensitive evidence-organization framework rather than a universal generic-quality optimizer.

5.7 Diagnostic Case Studies

Figure 3 diagnoses a failure mode rather than another best-case example. We compare against graph retrieval (no trust) because it is the closest controlled counterpart to AnchorRAG: it preserves the same evidence-graph and community-retrieval pipeline while removing the source-trust objective, so the contrast isolates the effect of trust-aware selection. On imm-01, which asks about immigration's effects on native wages and employment, AnchorRAG reaches

Table 4: Trust-sensitive evidence use in final answers on AllSides/Qbias (210 queries, seven topics). These metrics track whether retrieved high-trust evidence appears in generated citations and leading backbone claims. "graph ctrl." denotes graph retrieval (no trust). Asterisks follow Section 5 and indicate the least significant paired comparison against the six comparison methods.

Method	Cite Cov.	HT Cite.	Anchor Util.	Backbone Trust	Avg $S_{\text{trust}}@k$
vanilla rag [19]	0.3245	0.4061	0.4422	0.3937	0.7326
mmr rag [6]	0.3607	0.3681	0.6369	0.3412	0.7205
graph ctrl.	0.3457	0.4160	0.8997	0.3905	0.7290
R2AG [35]	0.3365	0.3871	0.4984	0.3946	0.7326
ReClaim [34]	**0.9344**	0.4245	0.4909	0.4209	0.7326
GRAG [15]	0.3943	0.3964	0.2914	0.3875	0.7318
AnchorRAG	0.6316	**0.6584****	**0.9180**	**0.6616****	**0.7635****

Table 5: External re-scoring of the same generated citations using MBFC factual reporting labels (210 queries, seven topics). MBFC HT Cite and MBFC Backbone HT are MBFC-label counterparts of HT Cite and Backbone Trust in Table 4; Avg Factual columns report mean MBFC factuality, and MBFC Map Cov. reports the fraction of citations whose sources are mapped to MBFC-rated outlets. Asterisks follow Section 5; † marks the borderline *MBFC HT Cite* cell against vanilla rag ($p = 0.078$). The final row reports the largest paired-comparison p-value for AnchorRAG against the six comparison methods.

Method	MBFC HT Cite	MBFC Avg Factual	MBFC Backbone HT	MBFC Backbone Avg Factual	MBFC Map Cov.
vanilla rag [19]	0.3790	3.2118	0.3464	3.1476	0.8894
mmr rag [6]	0.3283	3.1009	0.2672	2.9998	0.8749
graph ctrl.	0.3418	3.1461	0.2982	3.0788	0.8853
R2AG [35]	0.3342	3.1340	0.3280	3.1221	0.8948
ReClaim [34]	0.3519	3.1939	0.3581	3.2040	0.8985
GRAG [15]	0.3449	3.1448	0.3228	3.0792	0.8964
AnchorRAG	**0.4228**†	**3.3648****	**0.4140***	**3.3587****	**0.9308***
max pairwise p	*0.078*	< 0.001	*0.011*	< 0.001	*0.024*

Table 6: Answer-quality diagnostics for final generated answers. Values are GPT-4o (G) /Claude (C); metrics are reference-free.

Method	Grd. (G/C)	Supp. (G/C)	Util. (G/C)
vanilla [19]	**.797 / .694**	.852 / .845	**.786 / .677**
mmr [6]	.792 / .665	.840 / .834	.783 / .664
graph ctrl.	.792 / .667	**.865 / .862**	.783 / .666
R2AG [35]	.766 / .637	.698 / .687	.750 / .614
ReClaim [34]	.751 / .562	.700 / .579	.709 / .513
GRAG [15]	.751 / .658	.790 / .779	.734 / .617
AnchorRAG	.712 / .609	.772 / .770	.662 / .514

higher HT Cite/Citation Coverage/Backbone Trust (1.00/0.50/1.00) than graph retrieval (no trust) (0.67/0.23/0.67), but its planner treats most retrieved cards as context-only evidence about immigration politics, public perceptions, or adjacent visa debates rather than direct labor-market evidence. The resulting answer is therefore cautious and under-informative, with lower Groundedness/Utility (0.20/0.15 vs. 0.85/0.80). In contrast, graph retrieval (no trust) gives a more useful synthesis by using less trust-optimized but more directly responsive evidence. This case clarifies the main trade-off underlying Table 6: trust-aware evidence discipline can improve citation trust while reducing the usefulness of generic answers when the high-trust evidence is less directly responsive to the question.

Table 7: Diagnostic top-10 trust-advantaged query slice from the v5 210-query GPT-4o evaluation. Graph retrieval (no trust) is used as the closest controlled no-trust graph comparison and the strongest Support comparison. Queries are selected by the largest HT Cite improvement over graph retrieval (no trust) among cases where citation coverage improves, and support is preserved. Δ denotes AnchorRAG minus graph retrieval (no trust).

Topic	QID	Query Summary	Δ HT Cite (↑)	Δ Cite Cov. (↑)	$\text{Support}_{\text{Graph}}$	$\text{Support}_{AnchorRAG}$
Abortion	abo-21	waiting periods, counseling, and informed consent	1.0000	0.9000	1.0000	1.0000
Immigration	imm-25	second-generation immigrant integration outcomes	0.8571	0.4000	1.0000	1.0000
Gun Control	gun-05	concealed-carry laws and violent crime	0.8571	0.3333	1.0000	1.0000
Immigration	imm-20	immigration, population growth, and environmental impact	0.8571	0.2564	1.0000	1.0000
Voting Rights	vote-14	inactive-voter roll purges and disenfranchisement	0.8000	0.3654	0.8000	1.0000
Climate Change	clim-21	Paris Agreement costs and benefits	0.8000	0.2667	1.0000	1.0000
Economy and Jobs	eco-27	infrastructure investment and private-capital crowding out	0.7143	0.5000	0.0000	1.0000
Economy and Jobs	eco-24	housing prices: zoning restrictions vs speculative investment	0.7143	0.5000	0.0000	1.0000
Gun Control	gun-11	gun buyback programs and violence reduction	0.7143	0.2500	0.6667	1.0000
Gun Control	gun-08	school shootings: gun access vs mental health or security	0.7143	0.5000	0.8571	1.0000
Average	–	–	**0.8028**	**0.4272**	**0.7324**	**1.0000**

5.8 Ablation Study

We next examine which components of AnchorRAG are responsible for the observed gains, on the same seven-topic 210-query setting and the gpt-4o primary judge. Table 8 compares the full system

Failure case (imm-01): high-trust evidence is less directly responsive.
Query: What conflicting evidence exists regarding the effects of immigration on native wages and employment?

Graph retrieval (no trust) answer excerpt
"The effects of immigration on native wages and employment are subjects of considerable debate … restricting immigration, particularly through measures like suspending H-1B visas, is intended to protect native workers' job prospects. However, a study by the National Bureau of Economic Research challenges this notion … [Card 2]."

AnchorRAG answer excerpt
"The current evidence set lacks direct studies or data on the economic impacts of immigration on native wages and employment [Card 1] [Card 2] [Card 3] [Card 5] [Card 6] [Card 7] [Card 8]. … There is a lack of direct studies or data … making it difficult to draw definitive conclusions."

Diagnosis. AnchorRAG improves citation provenance but under-answers because the high-trust backbone is less direct for wage/employment mechanisms. Graph retrieval (no trust) gives a fuller answer with lower trust focus. *Metrics:* graph retrieval (no trust) vs. AnchorRAG gives Groundedness/Utility 0.85/0.80 vs. 0.20/0.15, but HT Cite/Citation Coverage/Backbone Trust 0.67/0.23/0.67 vs. 1.00/0.50/1.00.

Figure 3: Diagnostic failure case (imm-01): AnchorRAG improves citation trust but under-answers when the high-trust backbone evidence is less directly responsive to the question.

Table 8: Compact ablation results averaged across seven topics and 210 queries. Grd.=Groundedness, Supp.=Support, HT=High-Trust Citation Rate, BB=Backbone Trust, TW=Trust-Weighted Claim Support, Cov.=Citation Coverage.

Variant	Grd.	Supp.	HT	Util.	BB	TW	Claims	Cov.
vanilla [19]	0.734	0.645	0.350	0.677	0.348	0.492	1.75	0.198
wo trust	**0.749**	0.772	0.415	**0.708**	0.409	0.587	2.76	0.281
wo anchor	0.704	**0.802**	**0.603**	0.640	0.608	0.623	2.23	0.262
wo div.	0.745	0.704	0.530	0.695	0.531	0.553	2.05	0.221
AnchorRAG	0.732	0.793	0.601	0.682	**0.613**	**0.624**	**2.84**	**0.298**

with ablated variants and folds behavior diagnostics into the same table.

The ablation is an objective decomposition rather than a winner-take-all comparison: *wo trust* obtains the highest *Groundedness* and *Utility*, *wo anchor* obtains the highest sentence-level *Support*, while the full system is strongest on the backbone-oriented trust metrics and behavior diagnostics. Compared with *vanilla rag*, AnchorRAG improves *Support* by 0.1480, *HT Cite* by 0.2511, *Backbone Trust* by 0.2651, and *TW Claim Support* by 0.1320, with negligible differences on *Groundedness* (−0.0021) and *Utility* (+0.0052); compared with *wo trust*, the largest gain is in trust-sensitive use (HT Cite +0.1861, Backbone Trust +0.2040). Removing anchors yields slightly higher local support, but the gap to the full system is small (0.8020 vs. 0.7927), and AnchorRAG produces more evaluable claims, higher citation coverage, and the strongest backbone-oriented trust scores; we therefore interpret anchor guidance as shifting the generator toward more citation-bearing and trust-weighted backbone use rather than imposing a large support penalty. Removing diversity lowers support and citation coverage, suggesting that diversity prevents the evidence set from collapsing onto a narrow cluster. The multi-anchor variant does not improve over the single-anchor system on either support (0.7705 vs. 0.7927) or trust-weighted claim support (0.6139 vs. 0.6239), supporting the single-anchor design choice. Although each ablated variant wins on one metric family, the full system is strongest on the backbone-oriented trust metrics, consistent with the central claim above.

6 Discussion

Overall, our findings suggest that AnchorRAG is best understood not as a universal answer-quality optimizer but as a trust-sensitive evidence-organization framework for controversial QA. The empirical sequence first validates the retrieval-side community and source-reliability signals, then shows in the downstream transfer and MBFC analyses that higher-trust evidence becomes more visible in citations and answer backbones. Sections 5.5 and 5.6 together support the same interpretation: the trust-sensitive gains are stable, while generic answer-quality losses reflect a directional trade-off rather than universal dominance. The diagnostic case study and ablations further explain where this trade-off comes from.

The downstream trade-off is therefore not evidence against trust-aware retrieval itself. The graph retrieval (no trust) control and the *wo trust* ablation show that graph structure alone can preserve strong generic support, while source-trust discipline shifts the answer backbone toward higher-trust evidence that is sometimes less directly responsive. The remaining challenge is generation policy: high-trust evidence still needs to be converted into complete, directly supported answers. This is consistent with prior work treating truthfulness, reliability, and informativeness as distinct objectives rather than a single scalar notion of answer quality [20, 22, 24].

The component analysis supports this interpretation. Trust-aware retrieval most directly improves citation quality and backbone trust; community anchors provide a compact structural scaffold; and diversity helps prevent collapse into a single local cluster. The negative top-2-anchor result further suggests that anchor selection should provide a focused backbone rather than a larger set of competing center points.

A practical implication is that evidence cards should become more relation-aware. The current cards expose source, trust, relevance, community, and excerpt-level information, but they do not state how each card relates to its anchor. Adding labels such as *reinforces*, *contradicts*, or *elaborates* could help the generator move from local summarization toward structural synthesis.

Our study has several limitations. First, the source-reliability prior used by AnchorRAG is estimated by an LLM-based auditor; MBFC provides an external validation signal for matched sources, but does not replace human auditing of the full source set. We also report Claude-based robustness checks, but stronger cross-model and human validation would improve confidence. Second, the evidence graph is an approximation of discourse structure, not a gold-standard map of argumentative relations. Third, the prior-method baselines are controlled reproductions of each method's core mechanism inside our shared corpus, retrieval-budget, generator, and evaluation pipeline, while graph retrieval is an internal no-trust diagnostic control over the same article-level evidence graph; neither should be read as an exact rerun of external systems with their original code, prompts, training data, or model choices. Finally, the evaluation remains primarily LLM-judge-based and sentence-level, which can disadvantage cautious claims that are difficult to ground in a single local evidence span.

Future work should improve the source-trust component with explicit factuality resources or human-validated reliability signals, refine anchor selection for diffuse or heterogeneous communities, and develop generation policies that better convert high-trust retrieval into directly supportable claims. Broader validation in scientific, legal, and policy-oriented evidence synthesis would also test whether trust-aware retrieval and anchor-guided synthesis generalize beyond controversial news QA.

7 Conclusion

We proposed AnchorRAG, a framework that operationalizes a trust-sensitive evidence organization reformulation of controversial QA by combining trust-aware evidence retrieval with anchor-guided synthesis. At retrieval time, AnchorRAG jointly considers relevance, source trust, and discourse-community coverage; at generation time, structured evidence cards and one community anchor per retrieved community expose these signals to the planner and generator.

Experiments over seven AllSides/Qbias topics and 210 controversial queries show that AnchorRAG consistently improves trust-sensitive evidence use, including high-trust citation behavior, anchor utilization, retrieval trust, and backbone-trust usage. These gains demonstrate that retrieval-side trust signals can meaningfully influence downstream grounded synthesis. Although general answer-quality metrics that reward completeness and decisiveness show the expected trade-off, the pattern supports our central claim: controversial QA benefits from responsibly retrieving and using stronger evidence, not from maximizing generic fluency.

Overall, the findings suggest that trust-aware retrieval is a useful direction for grounded multi-perspective question answering, and they highlight the importance of evaluating not only whether a system retrieves relevant evidence, but also whether it retrieves trustworthy evidence and uses it responsibly in the final answer.

GenAI usage disclosure. GPT-4o serves as the source-reliability auditor, primary judge, and answer generator across all methods (temperature 0, fixed prompts); Claude-Sonnet-4-6 is a secondary judge for robustness checks on answer-quality metrics; `text-embedding-3-small` embeds documents and queries. Generative AI tools also assisted editing and consistency checking; all final content, experiments, analyses, and claims were reviewed and controlled by the authors. The multi-role use of GPT-4o (auditor, generator, judge) is mitigated by external MBFC validation of the source prior and Claude-based cross-judge checks on downstream metrics.

References

[1] Hunt Allcott and Matthew Gentzkow. 2017. Social media and fake news in the 2016 election. *Journal of economic perspectives* 31, 2 (2017), 211–236.

[2] Akari Asai, Zeqiu Wu, Yizhong Wang, Avi Sil, and Hannaneh Hajishirzi. 2024. Self-rag: Learning to retrieve, generate, and critique through self-reflection. In *International conference on learning representations*, Vol. 2024. 9112–9141.

[3] Ramy Baly, Georgi Karadzhov, Dimitar Alexandrov, James Glass, and Preslav Nakov. 2018. Predicting factuality of reporting and bias of news media sources. In *Proceedings of the 2018 conference on empirical methods in natural language processing*. 3528–3539.

[4] Vincent D Blondel, Jean-Loup Guillaume, Renaud Lambiotte, and Etienne Lefebvre. 2008. Fast unfolding of communities in large networks. *Journal of statistical mechanics: theory and experiment* 2008, 10 (2008), P10008.

[5] Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, Trevor Cai, Eliza Rutherford, Katie Millican, George Bm Van Den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, et al. 2022. Improving language models by retrieving from trillions of tokens. In *International conference on machine learning*. PMLR, 2206–2240.

[6] Jaime Carbonell and Jade Goldstein. 1998. The use of MMR, diversity-based reranking for reordering documents and producing summaries. In *Proceedings of the 21st annual international ACM SIGIR conference on Research and development in information retrieval*. 335–336.

[7] Danqi Chen, Adam Fisch, Jason Weston, and Antoine Bordes. 2017. Reading wikipedia to answer open-domain questions. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 1870–1879.

[8] Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, Dasha Metropolitansky, Robert Osazuwa Ness, and Jonathan Larson. 2024. From local to global: A graph rag approach to query-focused summarization. *arXiv preprint arXiv:2404.16130* (2024).

[9] Shahul Es, Jithin James, Luis Espinosa Anke, and Steven Schockaert. 2024. Ragas: Automated evaluation of retrieval augmented generation. In *Proceedings of the 18th conference of the european chapter of the association for computational linguistics: system demonstrations*. 150–158.

[10] Santo Fortunato. 2010. Community detection in graphs. *Physics reports* 486, 3-5 (2010), 75–174.

[11] Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. 2023. Enabling large language models to generate text with citations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. 6465–6488.

[12] Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Mingwei Chang. 2020. Retrieval augmented language model pre-training. In *International conference on machine learning*. PMLR, 3929–3938.

[13] Zoltan Gyongyi, Hector Garcia-Molina, and Jan Pedersen. 2004. Combating web spam with trustrank. In *Proceedings of the 30th international conference on very large data bases (VLDB)*.

[14] Fabian Haak and Philipp Schaer. 2023. Qbias-a dataset on media bias in search queries and query suggestions. In *Proceedings of the 15th ACM Web Science Conference 2023*. 239–244.

[15] Yuntong Hu, Zhihan Lei, Zheng Zhang, Bo Pan, Chen Ling, and Liang Zhao. 2025. Grag: Graph retrieval-augmented generation. In *Findings of the Association for Computational Linguistics: NAACL 2025*. 4145–4157.

[16] Gautier Izacard and Edouard Grave. 2021. Leveraging passage retrieval with generative models for open domain question answering. In *Proceedings of the 16th conference of the european chapter of the association for computational linguistics: main volume*. 874–880.

[17] Zhengbao Jiang, Frank F Xu, Luyu Gao, Zhiqing Sun, Qian Liu, Jane Dwivedi-Yu, Yiming Yang, Jamie Callan, and Graham Neubig. 2023. Active retrieval augmented generation. In *Proceedings of the 2023 conference on empirical methods in natural language processing*. 7969–7992.

[18] Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 conference on empirical methods in natural language processing (EMNLP)*. 6769–6781.

[19] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems* 33 (2020), 9459–9474.

[20] Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. Truthfulqa: Measuring how models mimic human falsehoods. In *Proceedings of the 60th annual meeting of the association for computational linguistics (volume 1: long papers)*. 3214–3252.

[21] Nelson F Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. Lost in the middle: How language models use long contexts. *Transactions of the association for computational linguistics* 12 (2024), 157–173.

[22] Ryan Liu, Theodore R Sumers, Ishita Dasgupta, and Thomas L Griffiths. 2024. How do large language models navigate conflicts between honesty and helpfulness? (2024), 31844–31865.

[23] Chaitanya Malaviya, Subin Lee, Sihao Chen, Elizabeth Sieber, Mark Yatskar, and Dan Roth. 2024. Expertqa: Expert-curated questions and attributed answers. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. 3025–3045.

[24] Jacob Menick, Maja Trebacz, Vladimir Mikulik, John Aslanides, Francis Song, Martin Chadwick, Mia Glaese, Susannah Young, Lucy Campbell-Gillingham, Geoffrey Irving, et al. 2022. Teaching language models to support answers with verified quotes. *arXiv preprint arXiv:2203.11147* (2022).

[25] Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen-tau Yih, Pang Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023. Factscore: Fine-grained atomic evaluation of factual precision in long form text generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. 12076–12100.

[26] Preslav Nakov, Jisun An, Haewoon Kwak, Muhammad Arslan Manzoor, Zain Muhammad Mujahid, and Husrev Taha Sencar. 2024. A survey on predicting the factuality and the bias of news media. In *Findings of the Association*

for Computational Linguistics: ACL 2024. 15947–15962.

[27] Mark EJ Newman. 2006. Modularity and community structure in networks. *Proceedings of the national academy of sciences* 103, 23 (2006), 8577–8582.

[28] Lawrence Page. 1998. The pagerank citation ranking: Bringing order to the web. Technical report. *Stanford Digital Library Technologies Project, 1998* (1998).

[29] Fabio Petroni, Aleksandra Piktus, Angela Fan, Patrick Lewis, Majid Yazdani, Nicola De Cao, James Thorne, Yacine Jernite, Vladimir Karpukhin, Jean Maillard, et al. 2021. KILT: a benchmark for knowledge intensive language tasks. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. 2523–2544.

[30] Hannah Rashkin, Eunsol Choi, Jin Yea Jang, Svitlana Volkova, and Yejin Choi. 2017. Truth of varying shades: Analyzing language in fake news and political fact-checking. In *Proceedings of the 2017 conference on empirical methods in natural language processing*. 2931–2937.

[31] Jon Saad-Falcon, Omar Khattab, Christopher Potts, and Matei Zaharia. 2024. Ares: An automated evaluation framework for retrieval-augmented generation systems. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. 338–354.

[32] Kurt Shuster, Spencer Poff, Moya Chen, Douwe Kiela, and Jason Weston. 2021. Retrieval augmentation reduces hallucination in conversation. In *Findings of the Association for Computational Linguistics: EMNLP 2021*. 3784–3803.

[33] Veselin Stoyanov, Claire Cardie, and Janyce Wiebe. 2005. Multi-perspective question answering using the OpQA corpus. In *Proceedings of Human Language Technology Conference and Conference on Empirical Methods in Natural Language Processing*. 923–930.

[34] Sirui Xia, Xintao Wang, Jiaqing Liang, Yifei Zhang, Weikang Zhou, Jiaji Deng, Fei Yu, and Yanghua Xiao. 2025. Ground every sentence: Improving retrieval-augmented llms with interleaved reference-claim generation. In *Findings of the Association for Computational Linguistics: NAACL 2025*. 969–988.

[35] Fuda Ye, Shuangyin Li, Yongqi Zhang, and Lei Chen. 2024. R2AG: Incorporating retrieval information into retrieval augmented generation. In *Findings of the Association for Computational Linguistics: EMNLP 2024*. 11584–11596.

[36] Yuheng Zha, Yichi Yang, Ruichen Li, and Zhiting Hu. 2023. AlignScore: Evaluating factual consistency with a unified alignment function. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 11328–11348.

[37] Xiangrong Zhu, Yuexiang Xie, Yi Liu, Yaliang Li, and Wei Hu. 2025. Knowledge graph-guided retrieval augmented generation. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. 8912–8924.